



LR20xx LoRa® 4th Generation Transceiver Performance

Table of Contents

1. List of Figures	2
2. List of Tables	3
3. Introduction	4
4. LoRa® Overview	5
4.1. Benefits	5
4.2. Different generations	6
5. Sensitivity improvements of the LR20xx compared to previous generations	7
6. Resistance to frequency error: Generation 1, 2, and 3 versus Generation 4	8
7. Resistance to Doppler shift and Doppler rate.	10
8. Interference mechanisms and LoRa® immunity.	11
9. Coding rate enhancement	12
9.1. Convolutional encoding	12
9.2. Resistance to fast fading.	12
9.3. Blocking performance	13
9.4. Burst interference performance	14
9.4.1. Test parameters	15
9.4.2. Test setup	15
9.4.3. Burst interference test results	16
10. Multi-SF detection	19
11. Channel activity detection (CAD).	22
11.1. Preamble-only CAD	22
11.2. Preamble-only vs symbol CAD comparison	23
11.3. Multi-SF CAD	23
11.4. Fast CAD	23
12. Key performance advantage for ARIB	25
13. Automated intra-packet frequency hopping	26
13.1. SetLoraHopping command	26
13.2. PA ramp-up / ramp-down (TX only)	27
13.3. Backward compatibility	27
13.4. Comparison: LR20xx vs SX127x (legacy)	28
14. Maximum and minimum payload size in LoRa® uplink	29
15. LoRa® address filtering.	30
15.1. Filter address structure	30
15.2. Configuration	30
15.3. Benefits	30
16. Summary	31
16.1. Compatibility matrix: SF5–SF6 with Generations 1, 2, and 3	31
16.2. LoRa® performance summary across generations.	32
17. Glossary	33
18. Revision History	35
Disclaimer	36

Chapter 1. List of Figures

[Figure 1.](#) PER as a Function of Frequency Offset

[Figure 2.](#) Fast Fading

[Figure 3.](#) Blocking Performance of LR20xx SF7 BW 125 kHz at 915MHz

[Figure 4.](#) Burst Interference Testing

[Figure 5.](#) Long Interleaver Test Setup

[Figure 6.](#) Test Packet Success Rate for BW 500 kHz SF7 160ms +50 dB Burst Interference

[Figure 7.](#) Test Packet Success Rate for BW 500 kHz SF10 160ms +50 dB Burst Interference

[Figure 8.](#) Fast CAD Threshold Adjustment Principle

[Figure 9.](#) Intra-Packet Frequency Hopping Diagram

Chapter 2. List of Tables

[Table 1.](#) SNR Improvement between two consecutive SFs

[Table 2.](#) Sensitivity of LoRa® demodulator for different SFs (BW=125 kHz, CR=4/5)

[Table 3.](#) Sensitivity Comparison Between Generations

[Table 4.](#) Preamble CAD vs Symbol CAD Comparison

[Table 5.](#) Hopping Command Parameters

[Table 6.](#) Hopping SPI Frame Format

[Table 7.](#) Comparison: LR20xx vs SX127x (Legacy)

[Table 8.](#) Maximum Payload Size in LoRa® Uplink

[Table 9.](#) LoRa® Radio Compatibility Details

[Table 10.](#) LoRa® Performance Summary Across Generations

Chapter 3. Introduction

LoRa® is a long-range communication technology developed by Semtech, based on chirp spread spectrum (CSS) modulation. Semtech launched the first generation of LoRa® chips, SX1272/76, in 2013. The second generation, SX1261/62/SX1280, and third generation, LR1110/1120/1121, further enhanced performance.

The LR2021 is the first chip in Semtech's LoRa Plus™ family, incorporating Generation 4 LoRa® IP technology. It provides single bill-of-materials compatibility across all output power values, an improved power amplifier efficiency, and a flexible packet engine supporting numerous modulation schemes. This enables PHY-layer compatibility with multiple protocols in addition to LoRaWAN®, such as Z-Wave, Thread, Sigfox, WM-Bus, Wi-SUN®, and Bluetooth® Low Energy.

This document details the performance improvements introduced in LR20xx LoRa® Generation 4 Transceivers relative to previous generations, including sensitivity, data rates, channel activity detection, and interference immunity. The LR20xx maintains backward compatibility with existing LoRaWAN deployments.

Chapter 4. LoRa® Overview

Several key parameters determine the characteristics of a LoRa® signal:

- **Spreading factor (SF):** The spreading factor controls the chirp rate and data transmission speed. Lower spreading factors produce faster chirps and higher data rates; higher spreading factors improve receive sensitivity and extend communication distance. LoRa® devices support SF5 to SF12, which are also supported by LoRaWAN since Regional Parameters RP002-1.0.5. The SFs are effectively orthogonal — they do not interfere with each other even when transmitted on the same RF frequency — increasing channel utilization.
- **Signal bandwidth (BW):** The bandwidth defines the frequency variation of the LoRa chirps and the spectrum bandwidth in kHz. LoRaWAN supports a subset of bandwidths (125, 250, or 500 kHz); sub-GHz LoRa devices support bandwidths from 7.81 kHz to 1000 kHz. LoRa Generation 2 added 2.4 GHz ISM band operation through SX1280, introducing bandwidths specific to 2.4 GHz such as 203, 406, and 812 kHz. Generations 3 and 4 support all available bandwidths independent of the configured RF frequency.
- **Coding rate (CR):** LoRa uses forward error correction (FEC) to provide robustness against environmental errors. The coding rate is configurable from 4/5 to 4/8.

4.1. Benefits

Long range

LoRa enables kilometres of communication distance. Range depends on the RF power emitted by the transmitting device and the receiver sensitivity. The link budget is expressed as:

$$\text{Link Budget (dB)} = \text{TX power} - \text{Sensitivity}$$

TX power correlates with current consumption and battery lifetime. In the 868 MHz band in Europe, TX power is limited to +14 dBm. In the corresponding sub-GHz bands in the United States, the maximum allowed output power for a LoRa® device is +30 dBm conducted.

LoRa® detects signals as low as 22 dB below the noise floor, achieving receiver sensitivities down to –141.5 dBm (LR20xx, Generation 4) and –141.0 dBm (LR11xx, Generation 3) at 868/915 MHz, SF12, BW 125 kHz. This gives Generation 4 LoRa® devices link budgets up to 155.5 dB (14 dBm – (–141.5 dBm)), and up to 168 dB in the United States at the higher permitted TX power. This link budget enabled LoRa® communication distance records exceeding 1300 km in September 2023.

Robust against the Doppler effect

The motion of one radio relative to another causes a shift in the received frequency due to the Doppler effect. LoRa® technology is robust against the Doppler effect and can be deployed in applications with high-speed moving assets, such as vehicle tracking, drones, and satellites. For more information, refer to application note [AN1200.80: LoRa® Modem Doppler Immunity](#).

Robust against interference

Due to its chirp spread spectrum nature, LoRa® is robust against in-channel and out-of-channel interference, making it suitable for dense or urban deployments. The signal resists both narrowband and wideband interference sources. For more information, refer to application notes [AN1200.22: LoRa® Modulation Basics](#) and [AN1200.30: Wi-Fi Immunity of LoRa® at 2.4 GHz](#).

Low cost

The frequency offset between transmitter and receiver can reach $\pm 25\%$ of the signal bandwidth, with moderate losses at band edges. This tolerance reduces the cost of LoRa® transmitters by eliminating the need for expensive TCXOs; crystals embedded in end devices need not meet extreme accuracy requirements.

4.2. Different generations

Four generations of LoRa® chips exist, each introducing new features and performance improvements:

- Generation 1 (Gen 1): SX127x
- Generation 2 (Gen 2): SX126x/SX128x and LLCC68
- Generation 3 (Gen 3): LR11xx
- Generation 4 (Gen 4): LR20xx

Gen 4 LoRa® radios communicate with previous generations provided they are configured with identical parameters (frequency, spreading factor, bandwidth, and coding rate).

Chapter 5. Sensitivity improvements of the LR20xx compared to previous generations

Radio sensitivity depends on the radio front-end noise figure (NF), the selected BW, and the demodulator signal-to-noise ratio (SNR), expressed as:

$$\text{Sensitivity} = \text{NF} + 10\log_{10}(\text{BW}) + \text{SNR (in dB for a given Packet Error Rate (PER))}$$

The SNR improvement between two successive SFs is expressed as:

$$10 \cdot \log\left(2 \cdot \frac{SF - 1}{SF}\right)$$

Table 1 summarizes the SNR improvements across consecutive SFs:

Table 1. SNR Improvement between two consecutive SFs

SF5 to SF6	SF6 to SF7	SF7 to SF8	SF8 to SF9	SF9 to SF10	SF10 to SF11	SF11 to SF12
2.2 dB	2.3 dB	2.4 dB	2.5 dB	2.5 dB	2.6 dB	2.6 dB

For example, SNR increases by 2.2 dB from SF5 to SF6. Table 2 shows the LoRa demodulator sensitivity for each SF:

Table 2. Sensitivity of LoRa® demodulator for different SFs (BW=125 kHz, CR=4/5)

SF5	SF6	SF7	SF8	SF9	SF10	SF11	SF12
-122 dBm	-125 dBm	-127.5 dBm	-130.5 dBm	-133 dBm	-136 dBm	-138.5 dBm	-141.5 dBm



Measurements in Table 2, “Sensitivity of LoRa® demodulator for different SFs (BW=125 kHz, CR=4/5)” were conducted with ±0.5 dB precision; sensitivity values therefore differ slightly from theoretical values.

The visible metric is sensitivity, as this directly impacts communication distance. Sensitivity integrates both the SNR characteristic of the LoRa® demodulator and the NF of the radio front end. Table 3 compares sensitivity across generations:

Table 3. Sensitivity Comparison Between Generations

Settings	SX127x	SX126x	LR11xx	LR20xx
SF7 BW 125 kHz	-124 dBm	-124 dBm	-127 dBm	-127.5 dBm
SF12 BW 125 kHz	-137 dBm	-137 dBm	-141 dBm	-141.5 dBm

With equivalent TX power, the LR20xx link budget exceeds that of the SX127x by 5 dB. In line-of-sight conditions, this approximately doubles communication distance at the same TX power, or alternatively supports a higher data rate over the same distance, reducing power consumption by at least half.

The LR20xx sensitivity improvements are achieved using a direct-tie RF implementation. Unlike previous generations, which required RF switch configurations to optimize receiver sensitivity separately from transmitter performance, the direct-tie architecture eliminates external RF switching components and avoids approximately 0.5 dB of component insertion loss. This delivers improved performance with reduced circuit complexity and lower BOM cost.

Chapter 6. Resistance to frequency error: Generation 1, 2, and 3 versus Generation 4

Frequency offset is a fundamental design constraint in radio system deployment. The LR20xx introduces a configurable setting to extend the frequency offset tolerance. When a transmitter and receiver operate at slightly different frequencies, communication reliability deteriorates rapidly. In previous LoRa® generations, frequency offset tolerance was limited to $\pm BW/4$, corresponding to ± 31.25 kHz for BW 125 kHz. Beyond this value, PER degrades rapidly, as shown in Figure 1, “PER as a Function of Frequency Offset”. The blue shaded region represents the BW/4 tolerance of ± 31.25 kHz. The LR20xx can be configured for a larger tolerance of $\pm BW/3$, corresponding to ± 41.7 kHz for BW 125 kHz, as indicated by the wider yellow shaded region. This enables selection of oscillators with wider frequency offset tolerance, which is beneficial for IoT applications where TCXO use is not desirable.

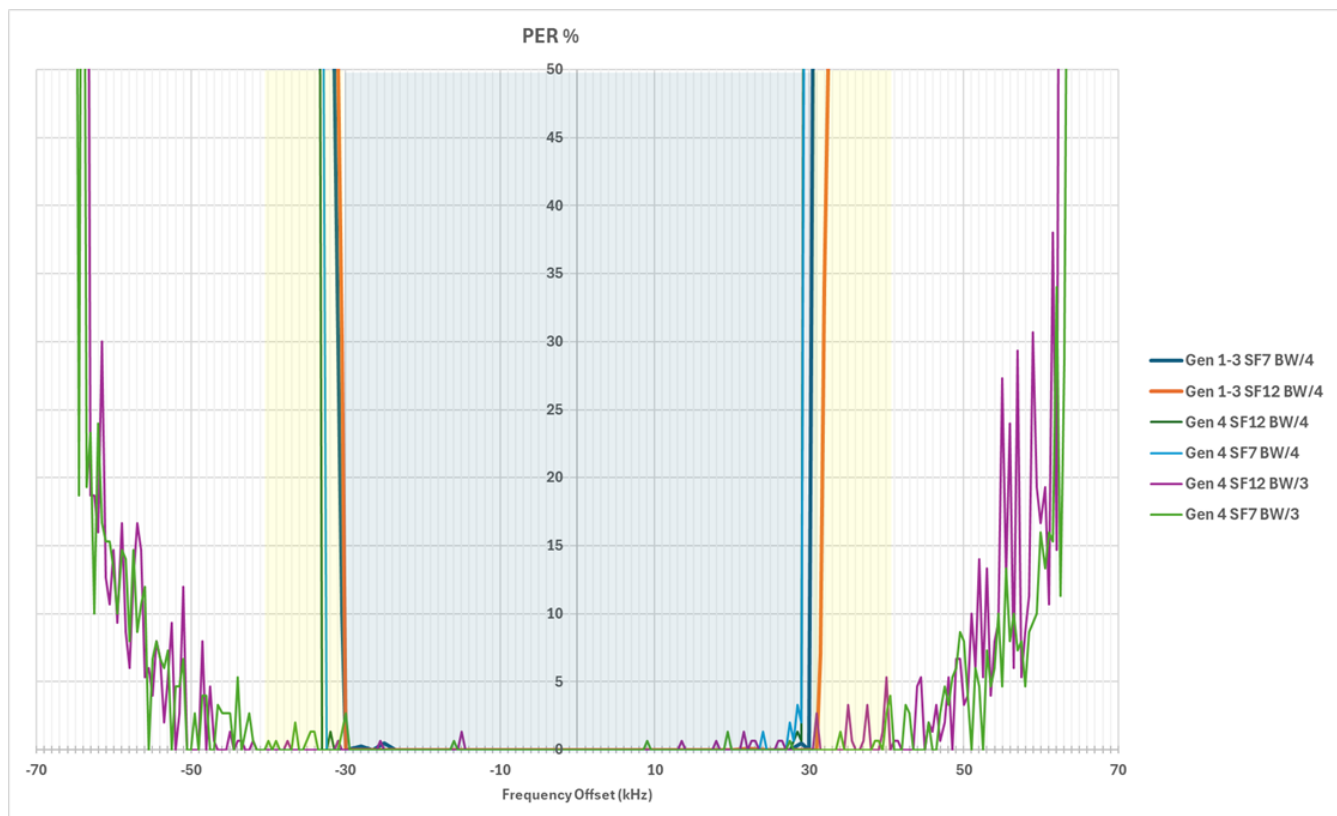


Figure 1. PER as a Function of Frequency Offset

High-precision crystal oscillators are required for reliable operation across a wide temperature range, often necessitating TCXOs on at least one radio in the link — typically the gateway in LoRaWAN deployments. For LoRa device-to-device communications, TCXO use is undesirable due to the additional 1–4 mA of continuous current consumption and increased BOM cost.

For more information, refer to:

- [AN1200.59: Selecting the Optimal Reference Clock](#)
- [AN1200.74: LoRa® Edge Clock Requirements](#)

The LR20xx extended frequency offset tolerance relaxes constraints on the 32 MHz crystal oscillator, enabling the use of lower-cost crystals.

Generation 4 supports two frequency error tolerance modes:

- Legacy (same as previous generations): $\pm BW/4$, for example ± 31.25 kHz for BW 125 kHz
- Extended: $\pm BW/3$, for example ± 41.7 kHz for BW 125 kHz

Legacy mode maintains $\pm BW/4$ tolerance, providing identical performance to previous generations and ensuring backward compatibility with existing deployments.

Extended mode provides 33% greater tolerance, relaxing oscillator requirements and enabling selection of lower-cost crystals — a key consideration in IoT deployments where device unit cost is a critical factor.

Both modes improve LoRa® system design flexibility.

The configuration of the level of robustness to frequency error is made via a register change, writing in bits 16 and 17 of register 0x00F30A2C, for example via the pseudo code command:

```
// Configure mode at address 0xF30A2C, bits [17:16]
WriteRegMemMask32(addr = 0xF30A2C, Mask = (3 << 16), Value = (mode << 16));
// where mode = 0 (legacy) or 1 (extended)
```

Chapter 7. Resistance to Doppler shift and Doppler rate

The Doppler effect presents a fundamental challenge in wireless communications when transmitters and receivers are in relative motion. This manifests in two forms: Doppler shift, the instantaneous change in received frequency due to relative speed; and Doppler rate, the rate of change of that frequency shift over time, typically caused by acceleration or changing velocity.

In traditional wireless systems, Doppler effects can cause communication failure when the frequency shift exceeds the receiver's tracking capability. LoRa® chirp spread spectrum modulation provides inherent robustness against these effects; the degree of tolerance has improved with each generation, enabling new categories of mobile applications. For more information, refer to application note [AN1200.80: LoRa® Modem Doppler Immunity](#).

Generation 1 and 2 receivers evaluate Doppler over the entire packet duration, depending on SF and BW, and typically support spreading factors up to SF8 in challenging Doppler environments.

Generation 3 improves frequency drift tolerance by evaluating Doppler over individual symbol periods rather than the full packet duration. Reducing the evaluation window from packet duration (potentially seconds) to symbol duration (typically milliseconds) significantly improves Doppler tolerance, enabling reliable communication up to SF10 or SF11 in demanding scenarios such as satellite communications.

Generation 4 further improves tracking range — the ability to follow and adjust to frequency changes over time — when LDRO is enabled. Built on the symbol-level evaluation established in Generation 3, Generation 4 maintains frequency lock more effectively throughout long symbol durations, improving communication reliability in dynamic frequency environments. This yields significant frequency drift tolerance for SF11 and SF12, where LDRO is typically enabled.

Chapter 8. Interference mechanisms and LoRa® immunity

In dense urban and industrial environments, multiple radio systems operating simultaneously create challenging interference conditions. Interference manifests in two primary forms: in-channel (co-channel) interference on the same frequency as the intended signal, and out-of-channel (adjacent channel) interference from nearby frequency bands. LoRa® chirp spread spectrum modulation provides inherent robustness against both types (see [AN1200.22: LoRa® Modulation Basics](#) and [AN1200.30: Wi-Fi Immunity of LoRa® at 2.4 GHz](#)).

Information bits are distributed across the entire LoRa® bandwidth rather than concentrated at specific frequencies, providing spectral diversity. Data is encoded over long symbol durations, enabling reconstruction even when portions of the signal are corrupted.

This makes LoRa® particularly effective against narrowband bursty interference, as the interference energy affects only a fraction of the total signal energy spread across frequency and time. The following sections describe the interference mitigation enhancements in the LR20xx Generation 4.

Chapter 9. Coding rate enhancement

LoRa® employs FEC through coding rates that add redundancy for error recovery. The coding rate expresses the ratio of useful information bits to total transmitted bits:

- CR 4/5: 4 information bits per 5 transmitted bits (20% redundancy)
- CR 4/6: 4 information bits per 6 transmitted bits (33% redundancy)
- CR 4/7: 4 information bits per 7 transmitted bits (43% redundancy)
- CR 4/8: 4 information bits per 8 transmitted bits (50% redundancy)

Higher coding rates provide greater error correction capability at the cost of longer transmission times and higher power consumption.

Introduced in Generation 3 (LR11xx) and enhanced in Generation 4 (LR20xx), long interleaving distributes coded bits across the entire LoRa® frame duration. This allows frame reconstruction even when consecutive LoRa symbols are corrupted, providing superior resilience against frequency-selective fading, burst interference longer than a single symbol duration, and fast fading.

The LR20xx adds Long Interleaver Convolutional codes 4/6 and 4/8 (CR8 and CR9 respectively). Convolutional coding spreads each data point across multiple adjacent coded symbols, creating overlapping protection that maintains signal integrity under severe interference conditions.

9.1. Convolutional encoding

Convolutional encoding is a form of forward error correction (FEC) in which the encoder has memory: each output symbol depends not only on the current input bit but also on a window of previous input bits. The encoder is typically implemented as a shift register and a set of modulo-2 adders (XOR gates) that form linear combinations of the stored bits. The ratio of input bits to output bits defines the code rate (e.g., 1/2 means one input bit produces two output bits); the number of shift-register stages is the constraint length, which determines how far back in the bit stream the encoder “remembers.”

In communications, convolutional codes are used to improve link reliability over noisy or fading channels. The transmitter encodes the data with the convolutional encoder before modulation; the receiver uses a decoder (e.g., Viterbi or sequential decoding) to exploit the redundancy and correct errors. Benefits include:

- **Error correction** – Redundancy allows the receiver to detect and correct a limited number of bit errors without retransmission.
- **Efficient decoding** – Viterbi decoding performs maximum-likelihood sequence estimation with manageable complexity for moderate constraint lengths.
- **Flexibility** – Puncturing (removing some encoded bits) allows higher code rates (e.g., 4/5, 4/6, 4/7, 4/8) from a single base encoder, trading redundancy for throughput.

Convolutional encoding is widely used in wireless standards (e.g., cellular 2G/3G, satellite, and many proprietary radios) and is often combined with other techniques such as interleaving and cyclic redundancy check (CRC) to improve performance under fading and burst errors.

9.2. Resistance to fast fading

Fast fading occurs when signal strength changes rapidly due to relative motion between transmitter and receiver, or changes in the propagation environment. Fast fading results from multiple concurrent environmental effects including multipath propagation (signals reaching the receiver via multiple paths), the Doppler effect (frequency shifts due to relative motion), and constructive/destructive interference from moving obstacles, rotating

machinery, and similar sources.



Figure 2. Fast Fading

The long interleaver in the LR11xx provides robustness against fast fading by distributing information across the LoRa® frame. However, losing a series of successive LoRa® symbols compromises reception even with the long interleaver enabled. The LR20xx convolutional coding schemes, CR8 (4/6) and CR9 (4/8), provide greater robustness against this effect because information within each symbol is spread across a window of successive symbols.

Convolutional coding can be affected by long error bursts; however, when combined with long interleaving, errors are distributed across the entire packet rather than concentrated in one region. Used together, the long interleaver and convolutional code provide strong robustness against fast fading and general interference effects.

CR8 and CR9 activate the convolutional coding feature. Like the legacy coding rates CR 4/5 to 4/8, they add redundancy to the LoRa® frame (4/6 for CR8 and 4/8 for CR9) and increase time on air.

9.3. Blocking performance

RF interference occurs both within the selected channel (in-band) and outside it (out-of-band). LoRa® has inherent robustness against out-of-channel interference, supporting strong interfering signals without degradation of link performance. These strong out-of-channel interferers are referred to as blockers or blocking signals. The blocking performance test applies a continuous unmodulated interfering signal at various frequency offsets from the wanted LoRa® signal, with blocker power progressively increased until PER reaches 1%. The test sets the wanted signal at sensitivity + 3 dB to reflect realistic operating conditions. This methodology characterizes receiver selectivity — the ability to discriminate between the desired signal and unwanted interference at different frequency separations.

Figure 3, “Blocking Performance of LR20xx SF7 BW 125 kHz at 915MHz” compares the blocking performance of the LR20xx (solid lines) and LR11xx (dotted lines) over a frequency offset range of -2 MHz to +2 MHz, using a continuous unmodulated blocking signal at SF7 BW 125 kHz at 915 MHz.

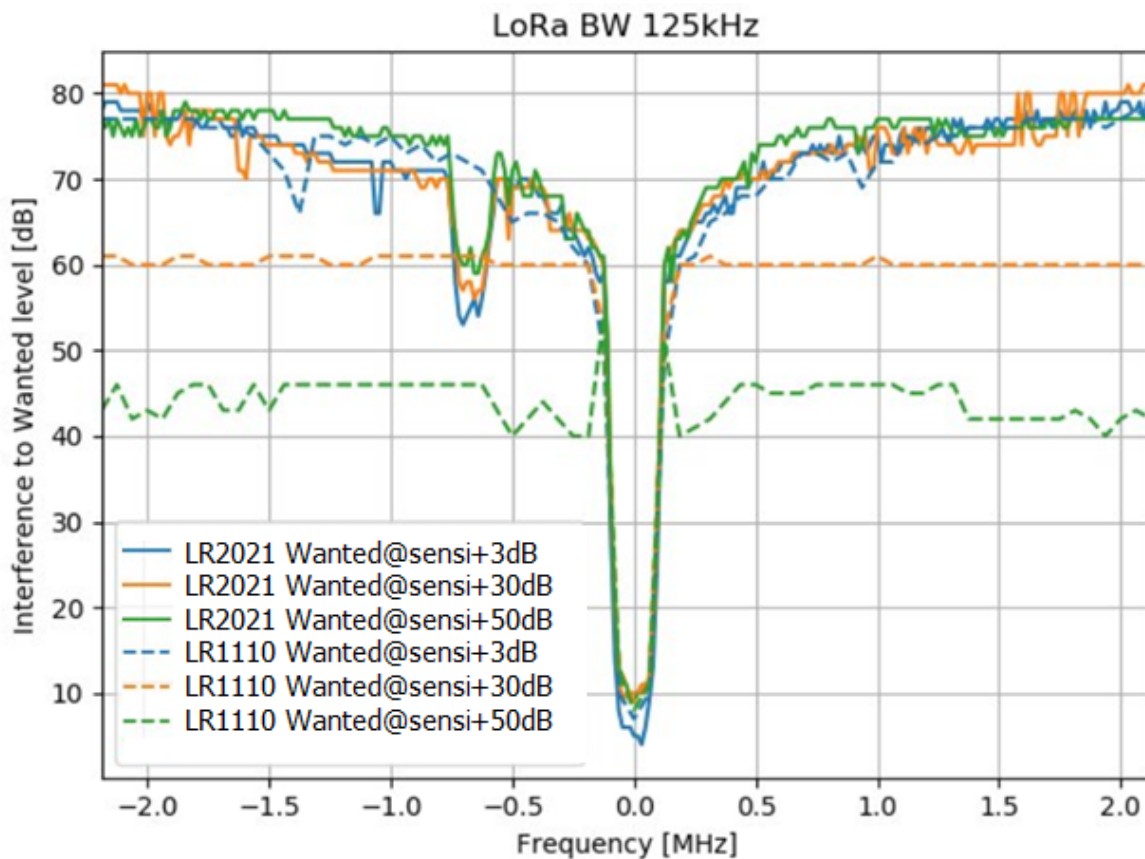


Figure 3. Blocking Performance of LR20xx SF7 BW 125 kHz at 915MHz

The LR20xx supports blocking signals at least 64 dB stronger than the wanted LoRa® signal at 1 MHz offset, and at least 76 dB stronger at 2 MHz offset, achieving comparable performance to the LR1110 for small wanted signals. Unlike the LR1110, the LR20xx maintains this performance when the wanted signal level is increased, demonstrating superior blocking performance near saturation.

Blocking performance is determined by the cumulative performance of the entire receive chain — from RF front-end components (low noise amplifier, anti-aliasing filter, mixer, automatic gain control) through digital signal processing blocks and the LoRa® demodulator. The LR20xx incorporates improvements throughout the receive chain that together deliver superior blocking immunity compared to previous generations, particularly in the presence of strong wanted and unwanted signals.

9.4. Burst interference performance

When blocking signals are intermittent, the interference is referred to as bursty interference. The burst interference performance test applies an interfering signal at various duty cycles overlapping in time with the wanted LoRa® signal. As shown in Figure 4, “Burst Interference Testing”, the burst interference signal is set N dB above the desired signal level, where N is typically 50 dB or more to ensure the burst interference is the dominant signal. Although the burst interference is significantly stronger than the LoRa® signal, it affects only a portion of the LoRa® signal. FEC allows the receiver to correct the resulting errors. The following sections compare the performance of the various FEC modes available in the LR20xx under representative burst interference conditions.

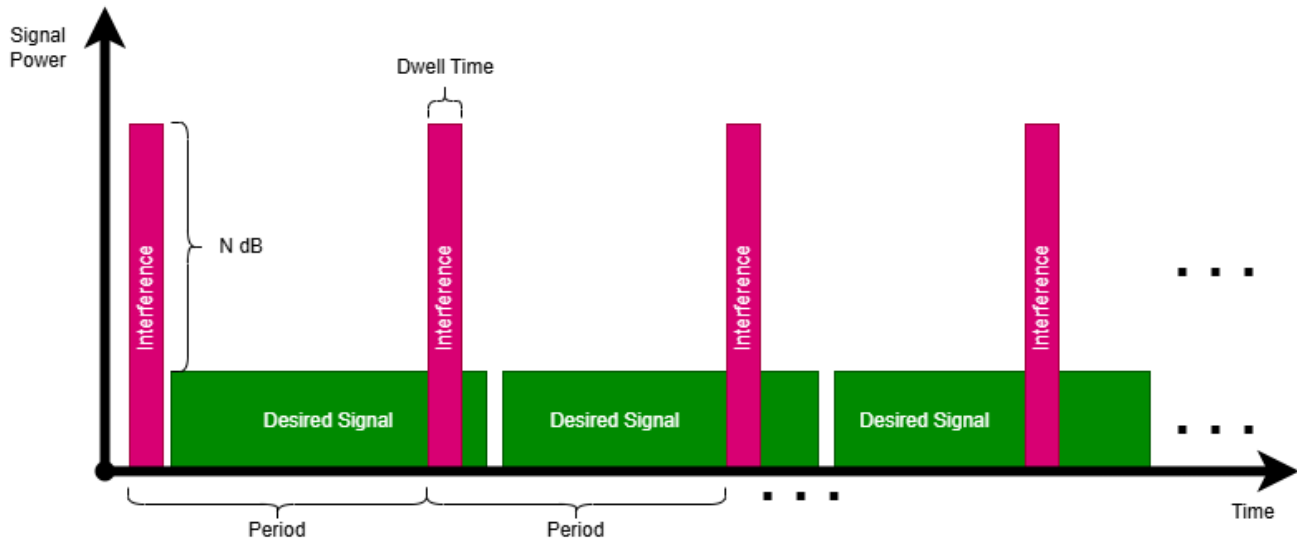


Figure 4. Burst Interference Testing

9.4.1. Test parameters

Based on Figure [Figure 4, “Burst Interference Testing”](#), duty cycle (DC) is defined as the percentage of time the burst interference signal is present within the wanted LoRa® signal, computed as $DC = \frac{DwellTime}{Period}$. The burst interference signal was selected to represent realistic in-band sub-GHz radio signals, well represented by Wi-SUN and other IEEE 802.15.4z compliant radio signals.

LoRa® Parameters (Desired Signal)

- BW: 500 kHz
- SF: 7, 10
- Payload length: 250 (SF7), 25 (SF10)
- Power level: 10 dBm
- Code Rate: Test variable (see note)
- Explicit header mode
- Preamble: 8 symbols
- Low Datarate Optimize: off
- IQ invert: off
- CRC: On

Burst Interference Parameters

- Modulation: 2-(G)FSK
- Bit Rate: 100 kbps
- Frequency Deviation: 50 kHz
- Power level: +10 dBm
- BT: 0.5
- Duty cycle: varies from 0% to 30%
- Data pattern: 000111



Attenuator adjusted so that desired signal measured 50 or 65 dBm below the interference signal level

9.4.2. Test setup

The burst interference signal is set 50 dB above the desired signal level to ensure it is the dominant signal.

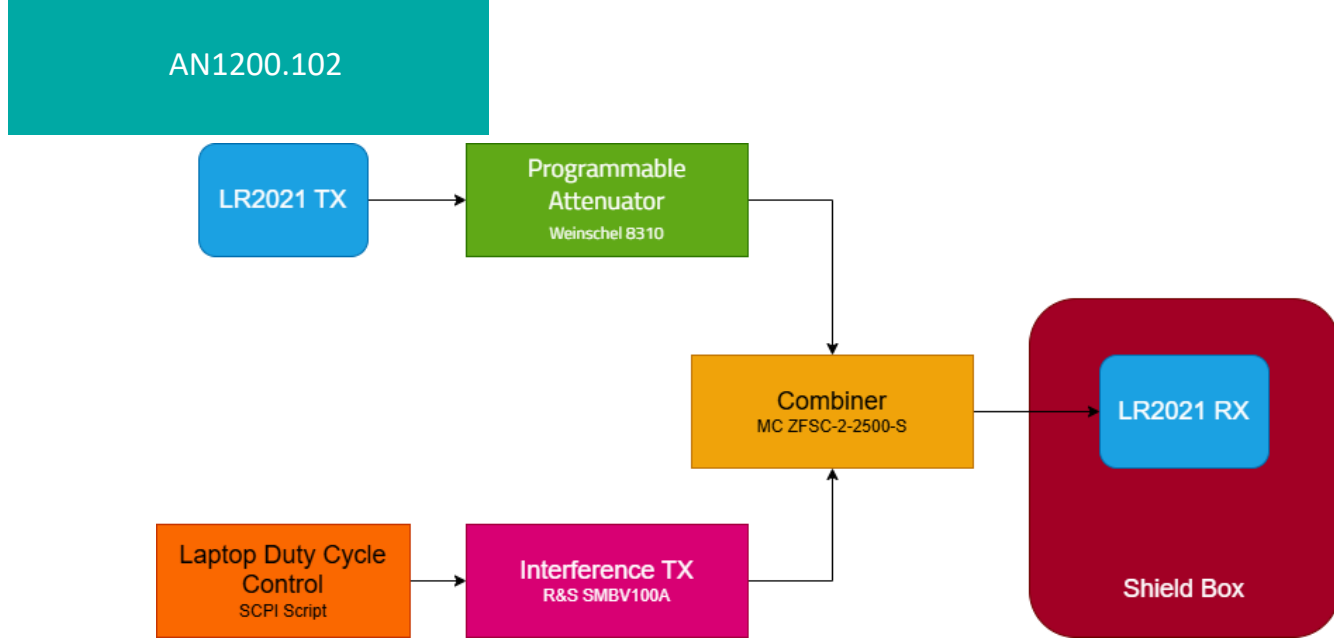


Figure 5. Long Interleaver Test Setup

Figure Figure 5, “Long Interleaver Test Setup” shows that the burst interference signal is shorter than the desired signal but at a much higher power level. The long interleaver (LI) or convolutional coding (CC) spreads the desired signal over the entire LoRa® frame duration, distributing errors across the packet rather than concentrating them in one region — enabling LI and CC to correct them. Additional coding rate redundancy also helps overcome short error bursts, though it is generally less effective than LI and CC.

9.4.3. Burst interference test results

Figure Figure 6, “Test Packet Success Rate for BW 500 kHz SF7 160ms +50 dB Burst Interference” shows the Packet Success Rate (PSR) as a function of duty cycle for BW 500 kHz, SF7, +50 dB burst interference with a 160 ms period. The interference duty cycle represents the on-time relative to the period (for example, 16 ms in 160 ms equals 10% DC). Packet length was adjusted to achieve approximately 100 ms time on air — approximately 250 bytes at BW 500 kHz with CR 4/5. Different CR values result in different packet lengths, adjusted to maintain approximately 100 ms air time. The test covered CR 4/5, CR 4/6, CR 4/7, CR 4/8, CR LI 4/5, CR LI 4/6, CR LI 4/8, CR CC 4/6, and CR CC 4/8.

Results show strong correlation across modes without LI or CC. LI 4/5 and LI 4/6 show a slight improvement below 8% DC, and LI 4/6 maintains improvement up to 15% DC. LI 4/8 shows slightly better improvement up to 18% DC. CC results are mixed: CC 4/6 performs slightly worse than LI 4/8, while CC 4/8 performs noticeably better than LI 4/8. This indicates that the additional redundancy of 4/8 relative to 4/6 can be more influential than the full-packet coding advantage of CC over LI.

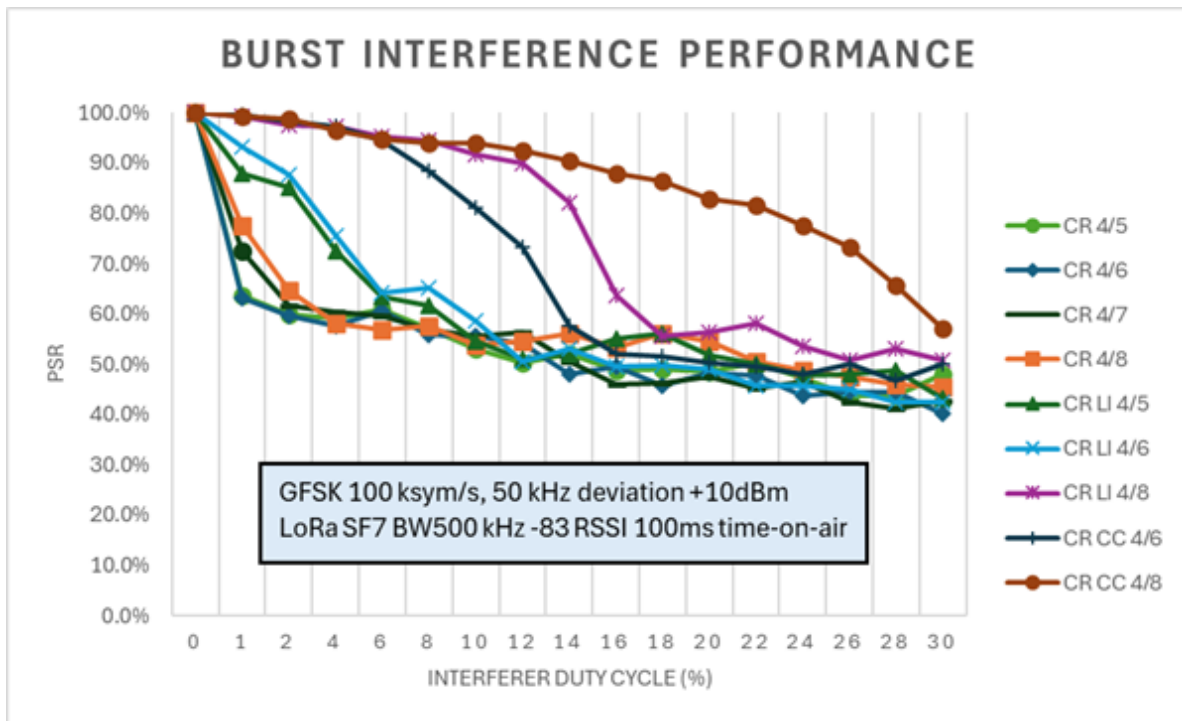


Figure 6. Test Packet Success Rate for BW 500 kHz SF7 160ms +50 dB Burst Interference

Figure Figure 7, “Test Packet Success Rate for BW 500 kHz SF10 160ms +50 dB Burst Interference” shows PSR as a function of duty cycle for BW 500 kHz, SF10, +50 dB burst interference with a 160 ms period. Packet length is 25 bytes, approximately 100 ms at BW 500 kHz with CR 4/5, adjusted per CR value to maintain approximately 100 ms air time. The test covered CR 4/5, CR 4/6, CR 4/7, CR 4/8, CR LI 4/5, CR LI 4/6, CR LI 4/8, CR CC 4/6, and CR CC 4/8. Results show strong correlation across modes without LI or CC, with stronger dependency on coding rate redundancy — CR 4/5 performs worse for both LI and non-LI modes. Differentiation between LI and CC is reduced compared to the SF7 results. Overall results are generally consistent across coding rates, suggesting that for +50 dB burst interference, the SF10 contribution due to longer symbol duration at the same DC is a significant performance factor.

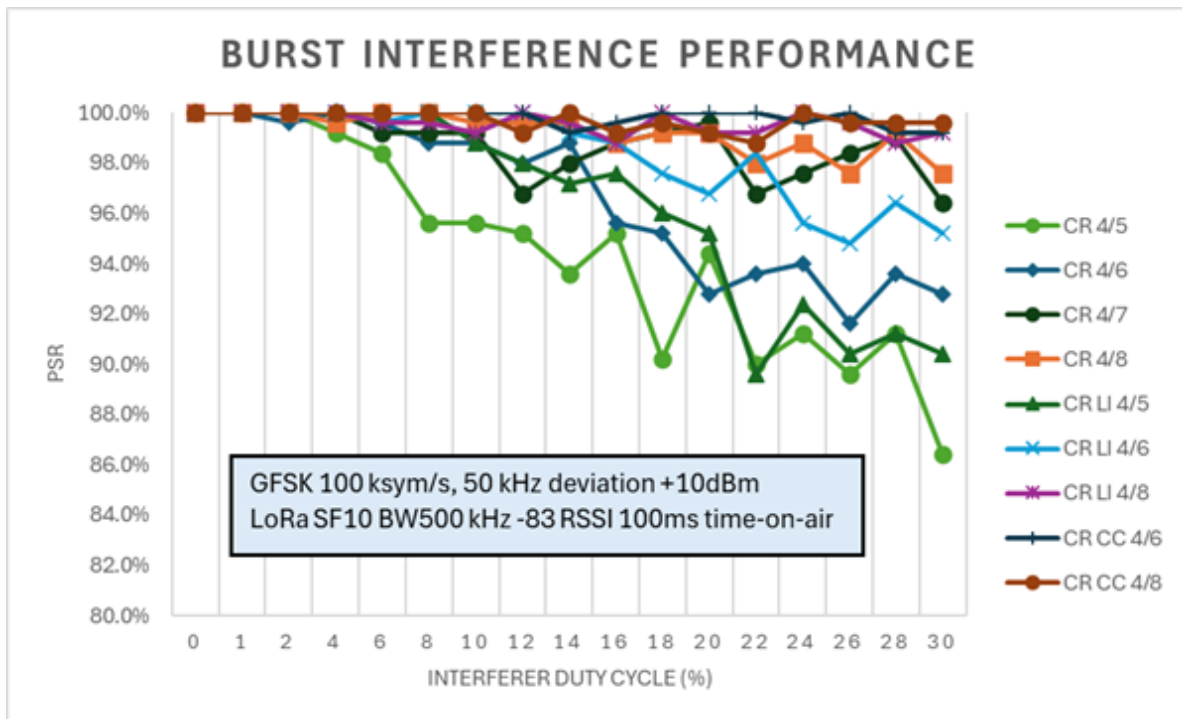


Figure 7. Test Packet Success Rate for BW 500 kHz SF10 160ms +50 dB Burst Interference

These results indicate that spreading factor and coding rate redundancy can have a greater impact on performance than the long interleaver or convolutional coding alone. A recommended approach to bursty interference is to increase spreading factor and coding rate redundancy first, then apply long interleaver and convolutional coding.

Chapter 10. Multi-SF detection

The LR20xx Generation 4 LoRa® IP introduces multi-SF detection, enabling LoRa® devices to handle mixed spreading factor environments. The multi-SF detection feature distinguishes between different SFs within a LoRa® packet and detects LoRa® symbols on multiple SFs in parallel (up to three side SFs). When RX is configured as CadExitMode, it automatically switches to RX mode to demodulate the detected SF. The system uses one main LoRa® detector plus up to three side detectors, each operating independently while sharing the same bandwidth and RF channel. In Generation 3, multi-SF was limited to a single side SF with a maximum value of 11.

Multi-SF detection operates under the following technical constraints:

- Spreading factor constraints
 - Up to four SFs can be detected simultaneously (main SF + three side SFs).
 - The smallest SF must be set on the main detector (via `SetLoraModulationParams`, described in the LoRa Commands section of the [LR2021 datasheet](#)). For CAD-MultiSF this rule is inverted: the largest SF is set on the main detector.
 - All configured SFs must be different; duplicate SFs are not permitted.
 - The difference between MaxSF and MinSF must be ≤ 4 .
- Bandwidth-dependent limitations
 - The sum of Detection Factors $\times$ BW must be strictly $< 32 \times 10^6$.
 - Detection factors for SF5–SF12: 10, 10, 12, 12, 14, 14, 16, 16.
 - Example: $(SF11 + SF12) \times BW1000$ is not allowed because $(16 + 16) \times 10^6 = 32 \times 10^6$ (not less than 32×10^6).

The following code example shows Multi-SF configuration for the LR20xx:

```
/**
 * @brief Configure Multi-SF for LoRaWAN gateway application
 *
 * A LoRaWAN gateway typically needs to receive on multiple channels and
 * multiple spreading factors. This example shows configuration for a single
 * channel with full SF coverage.
 *
 * Main SF: SF12 (highest sensitivity)
 * Side detectors: SF7, SF9, SF10
 * BW: 125 kHz
 * CR: 4/5
 * Detection limit: SF12+SF10+SF9+SF7 x BW125 results  $(16+14+14+12) \times 125e3 = 7e6 < 32e6$  (OK)
 *
 * For a complete gateway, you would replicate this for 8+ channels.
 */
void example_lorawan_gateway_channel(uint32_t freq_hz, uint8_t channel_id)
{
    smtc_rac_return_code_t status;
    uint8_t radio_id;

    /* Open radio for this channel */
    radio_id = smtc_rac_open_radio(SMTC_RAC_PRIORITY_NORMAL);

    /* Configure for LoRa packet type */
    status = smtc_rac_set_packet_type(radio_id, RAC_PKT_TYPE_LORA);
    status = smtc_rac_set_rf_freq(radio_id, freq_hz);
}
```

```

/* Main detector: SF12 (highest sensitivity) */
rac_lora_mod_params_t mod_params = {
    .sf      = RAC_LORA_SF12,
    .bw      = RAC_LORA_BW_125_KHZ, // LoRaWAN standard
    .cr      = RAC_LORA_CR_4_5,     // LoRaWAN coding rate
    .ldro    = 1,                   // Required for SF12 @ 125kHz
};
status = smtc_rac_set_lora_mod_params(radio_id, &mod_params);

/* LoRaWAN packet parameters */
rac_lora_pkt_params_t pkt_params = {
    .preamble_len_in_symb = 8,          // LoRaWAN preamble
    .header_type          = RAC_LORA_PKT_EXPLICIT,
    .payload_len          = 255,
    .crc_on               = true,        // LoRaWAN uses CRC
    .invert_iq            = false        // Uplink (normal IQ)
};
status = smtc_rac_set_lora_pkt_params(radio_id, &pkt_params);

/* Configure 3 side detectors to cover SF7-SF11
 * (SF12 is the main detector) */
rac_lora_side_det_config_t side_det[3] = {
    {
        /* SF7 - Fastest data rate, shortest range */
        .config    = 1,
        .sf        = RAC_LORA_SF7,
        .ldro      = 0,
        .invert_iq = 0
    },
    {
        /* SF9 - Mid-range performance */
        .config    = 1,
        .sf        = RAC_LORA_SF9,
        .ldro      = 0,
        .invert_iq = 0
    },
    {
        /* SF10 - Good sensitivity */
        .config    = 1,
        .sf        = RAC_LORA_SF10,
        .ldro      = 0,
        .invert_iq = 0
    }
};
status = smtc_rac_set_lora_side_det_config(radio_id, side_det, 3);

/* LoRaWAN public network syncword */
status = smtc_rac_set_lora_syncword(radio_id, 0x3444);
status = smtc_rac_set_lora_side_det_syncword(radio_id, 0x3444, 0x3444, 0x3444);

/* Start continuous RX */
status = smtc_rac_set_rx(radio_id, 0);
status = smtc_rac_submit_radio_transaction(radio_id);

printf("Gateway channel %u configured: %.3f MHz, SF7/9/10/12\n",
       channel_id, (float)freq_hz / 1e6);
}

```



After a successful reception, use the `GetLoraPacketStatus` command to determine which SF was demodulated. Calling `SetLoraModulationParams` disables all side SFs; reconfigure them after each call.



Configure the main SF using `SetLoraModulationParams` and the side SFs using `SetLoraSideDetConfig`, as described in the LoRa Commands section of the [LR2021 datasheet](#).

Chapter 11. Channel activity detection (CAD)

Wireless applications frequently need to detect radio traffic on air, either to identify a wanted signal (wake-on-radio) or to avoid collisions via Listen Before Talk (LBT) or Carrier Sense Multiple Access (CSMA). Because LoRa® can receive signals below the noise floor, RSSI-based measurements alone do not detect all LoRa® signals. The Channel Activity Detection (CAD) function addresses this limitation.

CAD detects any LoRa® communication packet on air for a given SF and bandwidth setting, including traffic within approximately 3 dB of the receiver sensitivity. The CAD process is designed to detect LoRa packets rapidly with minimal energy consumption. Depending on configuration, CAD can be automatically followed by TX (for LBT/CSMA) or RX (for wake-on-radio).

The LR20xx CAD implementation includes the following improvements over previous generations:

- CAD can be configured to detect LoRa® preamble symbols, payload symbols, or both.
- Multi-SF CAD performs parallel CAD across multiple SFs simultaneously.
- Fast CAD reduces power consumption when no LoRa® signal is present.

These improvements are described in the following sections.



The LR20xx can also perform CAD on GFSK frames using basic RSSI detection at a configurable threshold (see the SetCadParams command in the [LR2021 datasheet](#)). This feature is not covered in this document.

11.1. Preamble-only CAD

SX127x devices detect CAD on preamble symbols only. All other LoRa® products (SX126x, SX128x, LR11xx) detect any symbol in the entire packet, whether unmodulated (preamble) or modulated (payload).

The LR20xx CAD system provides selectable detection modes via the `pbl_any` parameter in the SetLoraCadParams command, described in the [LR2021 datasheet](#). When configured for preamble-only detection (`pbl_any=1`), the system focuses on unmodulated LoRa® preamble symbols, enabling faster transition to receive mode upon successful detection. With `pbl_any=1`, high-SNR LoRa® data symbols in the payload may still be detected, but detection is optimized for preamble symbols. The deterministic nature of preamble chirps reduces detection latency compared to full-packet symbol detection.

Table 4. Preamble CAD vs Symbol CAD Comparison

Parameter	Preamble-only CAD (<code>pbl_any=1</code>)	Full-Packet Symbol CAD (<code>pbl_any=0</code>)
Detection Target	Unmodulated preamble symbols only	All LoRa® symbols (preamble + payload)
Symbol Characteristics	Deterministic chirp pattern	Variable modulated data patterns
Ongoing Transmission Detection	Limited - only if preamble not missed	Complete — can detect midpacket
Detection Window	Packet start only	Entire packet duration
Optimal Use Cases	Wake-on-radio, CAD to Rx	Network monitoring, spectrum analysis
False Detection Probability	Lower in noise conditions	Higher potential at low SF/high SNR
Synchronization Advantage	Immediate packet duration knowledge	May require additional duration search

11.2. Preamble-only vs symbol CAD comparison

Preamble-only detection enables immediate transition to sync word search upon successful preamble detection, without analysis of the variable packet data that follows. Both preamble and modulated symbol detection consume less power than full RX mode by completing detection in a few symbols rather than receiving entire packets. This is particularly beneficial in battery-powered applications where CAD energy consumption directly impacts operational lifetime. For preamble-only detection, a more reliable approach than CAD is to use the `set_rx` command with an approximately six-symbol timeout, as CAD with `pb1_any=1` may occasionally detect LoRa symbols at high SNR.

CAD on LoRa symbols enables detection of packets already in progress. It also provides improved detection probability at high SNR due to the greater correlation energy available from modulated payload symbols.

11.3. Multi-SF CAD

Multi-SF CAD extends the parallel detection capabilities of the LR20xx to CAD operations, enabling simultaneous monitoring across multiple spreading factors within a single CAD cycle. The system uses the same detection architecture as multi-SF reception: one main detector plus up to three side detectors, each configured for independent spreading factor operation while sharing common bandwidth and channel parameters.

Configure Multi-SF CAD using the standard multi-SF setup sequence with CAD-specific parameter optimization, as described in the [LR2021 datasheet](#):

- LoRa® packet settings
 - `SetPktType()` — Configure packet type to LoRa mode.
 - `SetLoraModulationParams()` — Set main detector parameters (SF, bandwidth, coding rate).
 - `SetLoraPacketParams()` — Configure packet parameters and header settings.
- CAD settings
 - `SetLoraCadParams()` — Define CAD-specific behavior (detection duration, symbol targeting, Fast CAD parameters).
 - `SetLoraSideDetConfig()` — Enable and configure up to three side detectors for additional spreading factors.
 - `SetLoraCad()` — Initiate the configured multi-SF CAD operation.

11.4. Fast CAD

CAD detects a LoRa® signal from noise using as few symbols as possible, by comparing the received signal against a peak-to-noise ratio threshold. A threshold that is too high may discard weak signals, causing detection failures. A threshold that is too low may incorrectly detect noise as a LoRa® signal. The LR20xx introduces a variable threshold capability, Fast CAD, to overcome the fixed threshold limitations of previous generations.

Figure [Figure 8, “Fast CAD Threshold Adjustment Principle”](#) illustrates Fast CAD operation for five-symbol detection. Fast CAD progressively increases the peak-to-noise threshold for each successive symbol. If an actual LoRa packet is detected, the full detection process completes. If no packet is present, the process stops after one or two symbols rather than evaluating all five. The exact number of symbols depends on the configured CAD symbol count. Because Fast CAD terminates earlier than standard CAD when no signal is present, it reduces CAD power consumption and enables a more power-efficient LBT/CSMA process.

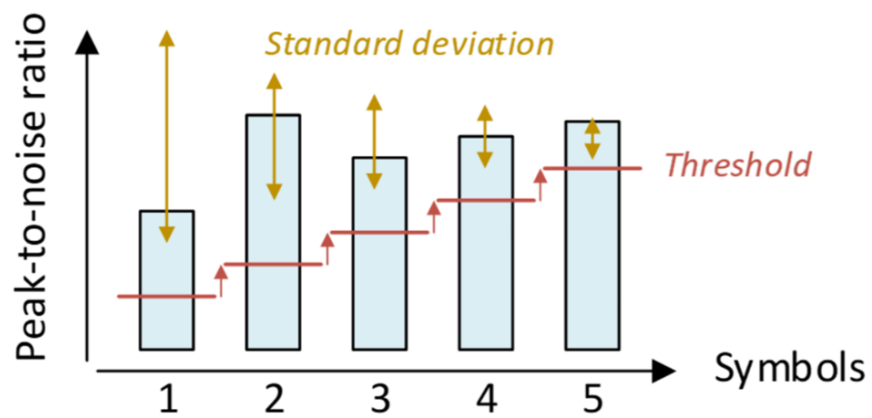


Figure 8. Fast CAD Threshold Adjustment Principle

Chapter 12. Key performance advantage for ARIB

ARIB (Association of Radio Industries and Businesses) is the Japanese standards organization that develops technical standards and regulations for radio equipment and telecommunications systems. ARIB establishes the requirements for wireless devices operating in Japan, performing an analogous role to the FCC in the United States and ETSI in Europe.

The LR20xx improved RF architecture eliminates the need to reduce transmission power when meeting ARIB requirements, and may reduce the need for expensive filters required in previous generations. This is due to:

- Improved phase noise performance (expected 3 dB margin at +14 dBm transmit power)
- Reduced out-of-band emissions (up to 9 dB higher transmit power without exceeding limits, relative to Generation 3)
- Better frequency offset tolerance ($\pm 33\%$ BW vs $\pm 25\%$ BW in previous generations)
- Enhanced sensitivity (-141.5 dBm at SF12/BW 125 kHz in sub-GHz)

Chapter 13. Automated intra-packet frequency hopping

The LR20xx intra-packet frequency hopping feature automatically changes the transmission frequency during a LoRa® packet using a frequency hopping sequence stored in device memory. The preamble and sync word are always transmitted on the initial frequency (channel 0). After the preamble and sync word are transmitted (TX) or received (RX), the transceiver counts `hop_period` LoRa® symbols and autonomously switches to the next frequency in the hopping table. This process repeats for each table entry, supporting up to 40 frequencies on the LR20xx. Intra-packet frequency hopping is useful in regions with dwell time limits on a single frequency, such as US915, enabling large packets to be transmitted without data fragmentation.

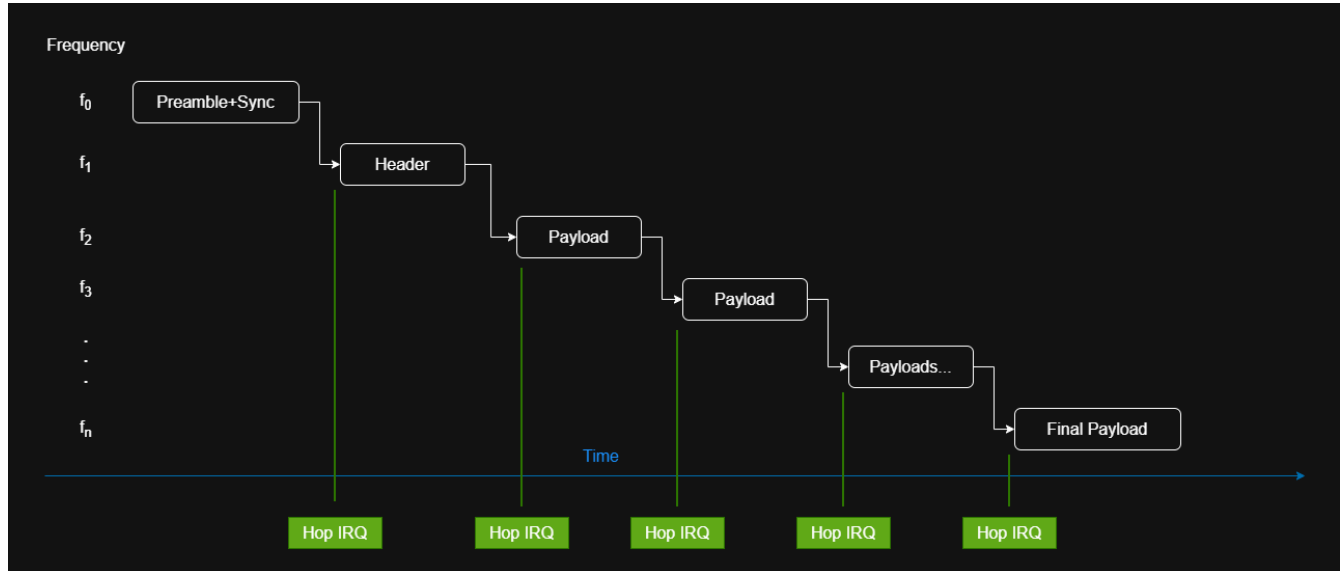


Figure 9. Intra-Packet Frequency Hopping Diagram

The Intra-Packet Frequency Hopping feature is enabled by setting the `hop_ctrl` parameter to 1 in the `SetLoraHopping` command and disabled by setting it to 0 (the `SetLoraHopping` command is described in the [LR2021 datasheet](#)).

13.1. SetLoraHopping command

This command configures intra-packet hopping in LoRa® mode. It accepts a control byte, a hop period, and an optional list of up to 40 frequency entries (4 bytes each, in the same format as `SetRfFrequency` — frequency in Hz, 32-bit unsigned).

Table 5. Hopping Command Parameters

Parameter	Width	Description
<code>hop_ctrl</code>	2 bits	0 = disabled, 1 = enable hopping. Others reserved.
<code>hop_period</code>	13 bits	Number of LoRa® symbols between frequency changes (after header).
<code>freq_hop[n]</code>	4 bytes each	Frequency in Hz for hop index n. Max 40 entries.



If `hop_ctrl` = 0, hopping is disabled and remaining parameters are ignored. When the frequency list is exhausted, the packet continues on the last frequency without interruption.

Table 6. Hopping SPI Frame Format

Byte	0	1	2	3	4	5	6	7	...
MOSI→	0x02	0x2C	hop_ctrl(1:0) in bits [7:6] RFU in bit[5:5] hop_period(12:8) in bits [4:0]	hop_period(7:0) in bits [7:0]	freq_hop1(31:24) in bits [31:24]	freq_hop1(23:16) in bits [23:16]	freq_hop1(15:8) in bits [15:8]	freq_hop1(7:0) in bits [7:0]	...
MISO←	Stat(15:8)	Stat(7:0)	IrqStat(31:24)	IrqStat(23:16)	IrqStat(15:8)	IrqStat(7:0)	0x00	0x00	...

Byte 2 packs hop_ctrl in bits [7:6], reserved bits [5:5], and the upper 5 bits of hop_period in bits [4:0]. Byte 3 carries the lower 8 bits of hop_period. Frequencies follow as big-endian 32-bit words.

Example - 3 Channel Hopping, Period = 20 Symbols

```
// Frequencies: 903.0 MHz, 904.6 MHz, 906.2 MHz
uint8_t cmd[] = {
    0x02, 0x2C, // Opcode: SetLoraHopping
    0x40 | ((20 >> 8) & 0x1F), // hop_ctrl=1 (bit7:6=01),
    (20 & 0xFF), // hop_period[12:8]
    // freq_hop[0] = 903000000 = 0x35D2AFC0
    0x35, 0xD2, 0xAF, 0xC0,
    // freq_hop[1] = 904600000 = 0x35EB19C0
    0x35, 0xEB, 0x19, 0xC0,
    // freq_hop[2] = 906200000 = 0x360383C0
    0x36, 0x03, 0x83, 0xC0,
};
spi_write(cmd, sizeof(cmd));
```

13.2. PA ramp-up / ramp-down (TX only)

During TX hops, the PA must be silenced before the PLL retunes to avoid spurious emissions. The ramp_down sequence disables the PA in a controlled ramp with the same duration as ramp_time from SetTxParams. After the frequency change, ramp_up re-enables the PA with the configured ramp time, managing PA supply, OCP/OVP, and SIMO DC-DC sequencing.

In RX mode, only the PLL retune occurs — no PA management is required. The retune is significantly faster, but its duration must be considered for small ramp_time values and large RF hops.

13.3. Backward compatibility

The intra-packet frequency hopping feature is backward compatible with legacy LoRa® devices, specifically the SX127x (Generation 1). Legacy LoRa® devices can receive and transmit LoRa® packets with intra-packet frequency hopping enabled.

The LR20xx can emulate the SX127x hopping behavior in which the host MCU reprogrammed the frequency on every FhssChangeChannel interrupt. To enable this legacy mode:

```
// Enable legacy hopping mode via register write
WriteRegMemMask32(addr=0xf30a24, mask=(1 << 18), value=(1 << 18));
```

In legacy mode, the LR20xx delays the first hop by one symbol to emulate the SX127x hopping behavior.

To disable legacy mode:

```
// Disable legacy hopping mode via register write
WriteRegMemMask32(addr=0xf30a24, mask=(1 << 18), value=0);
```

13.4. Comparison: LR20xx vs SX127x (legacy)

Table 7. Comparison: LR20xx vs SX127x (Legacy)

Feature	LR20xx	SX127x (Legacy)
Hopping control	Autonomous — frequency table loaded once via SetLoraHopping	Host MCU reprograms frequency on each FhssChangeChannel IRQ
Max hop channels	40 channels (FW table)	Unlimited (host-managed LUT)
Hop Period register	hop_period (13-bit, max 8191 symbols)	RegHopPeriod (8-bit, max 255 symbols)
Frequency change IRQ	LoRaTxRxHop (bit 12 of 32-bit IRQ status)	FhssChangeChannel (bit 1 of RegIrqFlags)
Host intervention per hop	None — FW handles PLL hop + PA ramp autonomously	Required — host must write Frf registers + clear IRQ
TX PA handling at hop	FW auto-executes ramp_down → PLL hop → ramp_up	Host must manage via mode transitions or FSTx → TX
PLL hop time	Comparable; ≤ 5 MHz steps skip VCO startup for faster settling	20–50 μs (depends on step size, per TS_HOP spec)
Channel readback	Host counts LoRaTxRxHop IRQ events	FhssPresentChannel in RegHopChannel[5:0]
SPI interface	Command-based SPI with full 32-bit frequency words	Register-based (8-bit Frf MSB/MID/LSB writes)
Header CRC fail recovery	Fires LoRaHeaderErr IRQ (bit 9); host decides recovery	Auto-returns to channel 0 and restarts preamble search
Legacy compatibility	Available via WriteRegMemMask32(0xf30a24)	N/A

Chapter 14. Maximum and minimum payload size in LoRa® uplink

The absolute maximum LoRa® uplink payload is 255 bytes. The coding rate, interleaver length, and CRC can reduce the effective maximum. The payload size refers to the length of the data field in the LoRa® packet.

Table 8. Maximum Payload Size in LoRa® Uplink

		LR11xx		LR20xx	
		crc on	crc off	crc on	crc off
Code Rate	Code Rate Index				
CR 0	0	255	255	255	255
CR 4/5	1				
CR 4/6	2				
CR 4/7	3				
CR 4/8 = CR 1/2	4				
CR LI 4/5	5	253 (1)	255 (1)	253	
CR LI 4/6	6				
CR LI 4/8	7				
CR LI CC 2/3	8	—	—	255 (2)	255 (2)
CR LI CC 1/2	9	—	—	252 (2)	254 (2)
(1) The minimum length is 8 bytes in Long Interleaver (LI) footnote of the LR1110 User Manual					
(2) The minimum length is 4 bytes for the LR20xx Convolutional Code (CC)					

Chapter 15. LoRa® address filtering

The LoRa® address filtering feature, introduced in Generation 4 (LR20xx), enables the device to discard LoRa® packets not addressed to it. This reduces host processing load and power consumption in dense deployments by rejecting unwanted packets early in the receive chain, before full payload demodulation.

The address filter operates on bytes extracted from the received LoRa® payload. A configurable address byte sequence is compared against received address bytes to accept or reject each packet. Packets that fail the filter are discarded and raise the `AddrError` IRQ, reducing host wake-ups and eliminating the overhead of processing unwanted packets.

15.1. Filter address structure

Configure the LoRa address filter value using the `SetLoraAddr` command, as described in the [LR2021 datasheet](#). The value contains up to eight bytes, matching between one and eight LoRa address values. The filter supports full 8-byte address matching or partial address matching.

A match occurs when the address bytes equal the received bytes at the specified byte position. On a match, the packet is accepted and reception continues. On a mismatch, the packet is rejected, RX is stopped, and an IRQ is raised.

15.2. Configuration

- Configure the filter table using `SetLoraAddr` before entering RX mode.
- The filter applies to a designated byte position in the LoRa payload.
- When enabled, packets that do not match any accept rule are discarded and the modem raises the `AddrError` IRQ.
- Disable address filtering by setting `addr_comp_len` to 0 in the `SetLoraAddr` command.



The exact byte position used for filtering, the encoding of the action field, and the behavior when no entries match may depend on the packet format (explicit/implicit header). Refer to the *LoRa Packet Handler* and *SetLoraAddress* sections of the [LR2021 datasheet](#) for definitive specifications.

15.3. Benefits

- Reduces host CPU wake-ups and processing for packets addressed to other nodes.
- Lowers power consumption in dense networks where many received packets are not intended for the device.
- Power consumption savings depend on packet size and transmit configuration (SF, BW, CR); however, power consumption with filtering enabled is always lower than without.
- Complements wake-on-radio and CAD by filtering at the modem level before full packet reception (for example, LoRaWAN® Relay).

Chapter 16. Summary

The LR20xx offers the following improvements over previous generations:

- Improved radio sensitivity.
 - With equivalent TX power, the LR20xx link budget exceeds that of the SX127x by 5 dB.
- New sub-GHz BW setting of 1000 kHz, enabling a maximum data rate of 125 kbps at SF5.
- Greater frequency offset tolerance.
 - Relaxes constraints on the 32 MHz crystal oscillator, enabling lower-cost crystals.
- Improved resistance to Doppler shift and Doppler rate.
 - Enhanced tracking range when LDRO is enabled, providing significant frequency drift tolerance at SF11 and SF12.
- Increased resistance to bursty interference and fast fading.
 - Convolutional codes CR8 and CR9 improve robustness against interferers and fast fading.
- New direct-tie architecture.
 - Eliminates external RF switching components.
 - Avoids approximately 0.5 dB component insertion loss.
- Improved blocking performance.
 - The LR20xx supports blocking signals at least 64 dB stronger than the wanted LoRa® signal at 1 MHz offset and at least 76 dB stronger at 2 MHz offset.
- Improved phase noise performance.
- Reduced out-of-band emissions.
- ARIB compliance without reduction in TX power.
- Improved CAD implementation.
 - CAD can be configured to detect LoRa® preamble symbols, payload symbols, or both.
 - Multi-SF CAD monitors LoRa® symbols on multiple SFs in parallel.
 - Fast CAD reduces power consumption when no LoRa® signal is detected.
- Automated intra-packet frequency hopping, backward compatible with legacy LoRa® devices.
- LoRa® address filtering.

16.1. Compatibility matrix: SF5–SF6 with Generations 1, 2, and 3

LoRa® radios communicate with each other provided they are configured with identical parameters (frequency, spreading factor, bandwidth, coding rate, LoRa® sync word, and chirp polarity). This enables adoption of new generations without requiring changes to application code or the air protocol. Table 7 summarizes LoRa® radio compatibility details:

Table 9. LoRa® Radio Compatibility Details

Radio Generation	Gen 1	Gen 2	Gen 3 and 4
Compatibility	- Supports SF6 with implicit header and special interleaving scheme - Does not support SF5	- Supports SF5 and SF6 with implicit and explicit header and standard interleaving scheme - Communication between Gen 1 and Gen 2 at SF6 is not possible	- Supports SF5 and SF6 (including legacy SF6 of Gen 1) - Supports both formats with a static configuration bit to communicate with Gen 1 at SF6 implicit header ^[1] - Compatibility mode with syncword format used in Gen 1 (Gen 4 only)

16.2. LoRa® performance summary across generations

Table 10. LoRa® Performance Summary Across Generations

Feature	SX127x (Gen 1)	SX1261/2 (Gen 2)	LR11xx (Gen 3)	LR20xx (Gen 4)
Sensitivity SF7	–124 dBm	–124 dBm	–127 dBm	–127.5 dBm
Sensitivity SF12	–137 dBm	–137 dBm	–141 dBm	–141.5 dBm
Max data rate	~62.5 kbps	~62.5 kbps	~62.5 kbps	125 kbps
Frequency offset tolerance	±BW/4	±BW/4	±BW/4	±BW/3 (extended mode)
CAD capabilities	Preamble only	Standard	Standard	Multi-SF + Fast CAD
Doppler tolerance	Packet-level	Packet-level	Symbol-level	Enhanced symbol + LDRO
Coding rates	CR 4/5 to 4/8	CR 4/5 to 4/8 + Long	CR 4/5 to 4/8 + Long	CR 4/5 to 4/8 + Long + Convolutional
Blocking immunity	Standard	Standard	Enhanced	64–76 dB @ 1–2 MHz offset

[1] Please refer to the LR20xx driver on github.com/Lora-net for an implementation example

Chapter 17. Glossary

Term	Description
ARIB	Association of Radio Industries and Businesses
BW	Bandwidth
CAD	Channel Activity Detection
CC	Convolutional Code
CR	Coding Rate
CSMA	Carrier Sense Multiple Access
CSS	Chirp Spread Spectrum
DC	Duty Cycle
FEC	Forward Error Correction
GFSK	Gaussian Frequency Shift Keying
IoT	Internet of Things
LBT	Listen Before Talk
LDRO	Low Data Rate Optimization
LI	Long Interleaver
LoRa®	Long Range. The LoRa mark is a registered trademark of Semtech Corporation.
LoRaWAN®	LoRa Wide Area Network
NF	Noise Figure
OCP	Over Current Protection
OVP	Over Voltage Protection
PA	Power Amplifier
PER	Packet Error Rate
PHY	Physical Layer
PLL	Phase Locked Loop
PSR	Packet Success Rate
RF	Radio Frequency
RSSI	Received Signal Strength Indicator
RX	Reception
SF	Spreading Factor
SNR	Signal-to-Noise Ratio
TCXO	Temperature Compensated Crystal Oscillator
TX	Transmission

Term	Description
VCO	Voltage Controlled Oscillator

Chapter 18. Revision History

Version	ECO	Date	Changes and/or Modifications
1.0	076680	October 2025	Preliminary Version
1.1	079168	June 2026	Add Intra-Packet Frequency Hopping description, expanded coding rate details with enhanced burst interference resistance, and added LoRa® address filtering description. Additionally there were several clarifications and corrections and editorial changes to improve the document clarity.

Disclaimer



Information relating to this product and the application or design described herein is believed to be reliable, however such information is provided as a guide only and Semtech assumes no liability for any errors in this document, or for the application or design described herein.

Semtech reserves the right to make changes to the product or this document at any time without notice. Buyers should obtain the latest relevant information before placing orders and should verify that such information is current and complete. Semtech warrants performance of its products to the specifications applicable at the time of sale, and all sales are made in accordance with Semtech's standard terms and conditions of sale.

SEMTECH PRODUCTS ARE NOT DESIGNED, INTENDED, AUTHORIZED OR WARRANTED TO BE SUITABLE FOR USE IN LIFE-SUPPORT APPLICATIONS, DEVICES OR SYSTEMS, OR IN NUCLEAR APPLICATIONS IN WHICH THE FAILURE COULD BE REASONABLY EXPECTED TO RESULT IN PERSONAL INJURY, LOSS OF LIFE OR SEVERE PROPERTY OR ENVIRONMENTAL DAMAGE. INCLUSION OF SEMTECH PRODUCTS IN SUCH APPLICATIONS IS UNDERSTOOD TO BE UNDERTAKEN SOLELY AT THE CUSTOMER'S OWN RISK. Should a customer purchase or use Semtech products for any such unauthorized application, the customer shall indemnify and hold Semtech and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs damages and attorney fees which could arise.

The Semtech name and logo are registered trademarks of the Semtech Corporation. All other trademarks and trade names mentioned may be marks and names of Semtech or their respective companies. Semtech reserves the right to make changes to, or discontinue any products described in this document without further notice. Semtech makes no warranty, representation or guarantee, express or implied, regarding the suitability of its products for any particular purpose. All rights reserved.

© Semtech 2026